In this traditional approach with server-side rendering, React renders components to a string of HTML and sends that HTML string to the client. Once the HTML DOM is presented, React reruns on the client side, operating over the same rendered HTML and adding interactive elements. This method allows for faster initial page loads compared to solely client-side rendering, but it still requires a hydration step to make the page fully interactive.

RSCs, however, introduce a new way of rendering. They allow specific parts of a React application to be rendered on the server. Figure 4.3 demonstrates how the React component tree is rendered using RSCs.

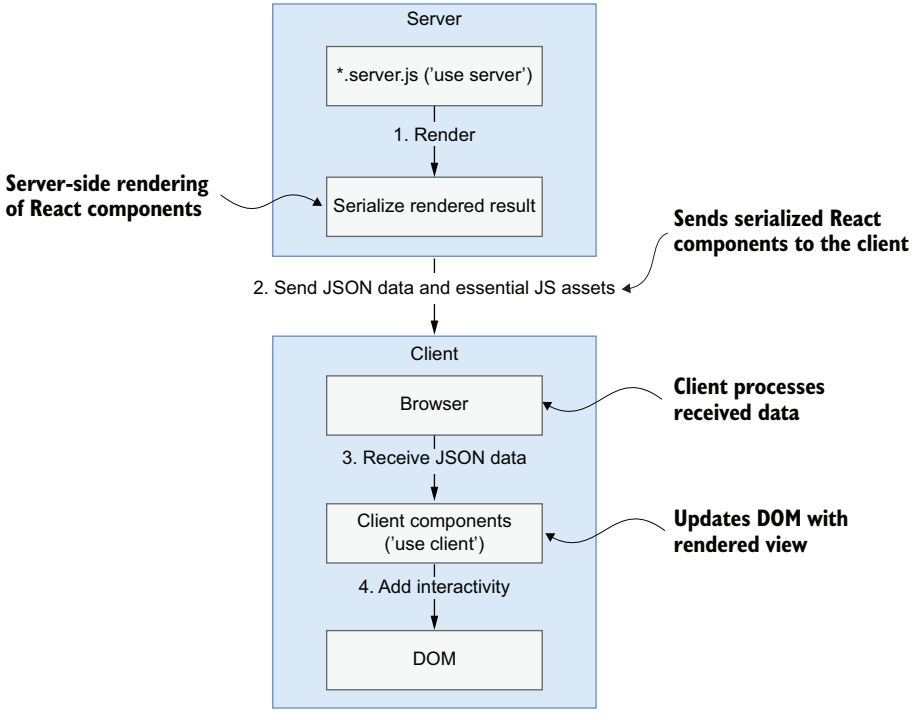


Figure 4.3 RSCs allow specific parts of a React application to be rendered on the server. The components can be streamed to the client incrementally, allowing the client to begin rendering parts of the UI before the entire page is ready.

In this new model, there is no hydration process for server components. Instead, these components are rendered on the server, and their output is sent to the client along with a lightweight JavaScript runtime. Client components are then rendered on the client and integrated into the DOM as HTML tags, adding interactivity where needed.